Biotechnology and genetic modification

Original exam-style questions · 20 questions

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Question 1

core · 1 mark · 0610_T21_P1_20261018_Q1

Why are bacteria useful for producing large quantities of a useful protein?

A They contain no genetic information.

B They can reproduce rapidly and make complex molecules.

C They never need nutrients.

D Every bacterium is a human body cell.

Question 2

core · 1 mark · 0610_T21_P1_20261018_Q2

Which yeast product makes bread dough rise?

A lactic acid

B oxygen

C cellulose

D carbon dioxide

Question 3

core · 1 mark · 0610_T21_P1_20261018_Q3

Which product of anaerobic respiration in yeast is used as a biofuel?

A ethanol

B urea

C starch

D oxygen

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Question 4

core · 1 mark · 0610_T21_P1_20261018_Q4

Why is pectinase added during some fruit-juice production?

A to convert all juice into protein

B to sterilise the juice by boiling

C to break down pectin and help release juice

D to turn sugar into oxygen

Question 5

core · 1 mark · 0610_T21_P1_20261018_Q5

Which comparison best tests the effect of enzymes in washing powder?

A different stains washed at different temperatures with different powders

B one fabric washed and another left completely dry

C different fabrics washed for different durations

D equal stained fabrics washed under identical conditions with enzyme and enzyme-free powders

Question 6

supplement · 1 mark · 0610_T21_P1_20261018_Q6

Which enzyme is used to make lactose-free milk?

A pectinase

B lactase

C lipase only

D amylase only

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Question 7

supplement · 1 mark · 0610_T21_P1_20261018_Q7

Why is cooling often needed in an aerobic industrial fermenter?

A Microorganisms always absorb all surrounding heat.

B Cooling removes all nutrients.

C Respiration by microorganisms releases energy that can raise temperature.

D Every enzyme works best at boiling temperature.

Question 8

core · 1 mark · 0610_T21_P1_20261018_Q8

Which action is genetic modification?

A inserting an individual gene into an organism's genetic material

B choosing two high-yield parents to breed without changing individual genes directly

C feeding an animal a more nutritious diet

D training an animal to respond to a signal

Question 9

core · 1 mark · 0610_T21_P1_20261018_Q9

A crop receives a gene that makes it resistant to an insect pest. What is the intended advantage?

A a guarantee that every insect species becomes extinct

B less crop damage from that pest

C no further need for water

D automatic resistance to every disease

Question 10

supplement · 1 mark · 0610_T21_P1_20261018_Q10

Which enzyme joins a human gene into a cut bacterial plasmid?

A pectinase

B lactase

C DNA ligase

D protease

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Question 11

core · 4 marks · 0610_T21_P1_20261018_Q11

A company uses bacteria to produce a medically useful protein.

(a) Explain two biological features that make bacteria suitable. [2]

(b) Explain how these features help production on a large scale. [2]

Question 12

supplement · 4 marks · 0610_T21_P1_20261018_Q12

Researchers consider bacteria rather than large mammals for a genetic-modification project.

(a) Explain the usefulness of plasmids. [2]

(b) State two reasons bacterial manipulation often raises fewer ethical concerns than using large mammals. [2]

Question 13

core · 4 marks · 0610_T21_P1_20261018_Q13

Yeast is used in both bread-making and biofuel production.

(a) State the two main products of anaerobic respiration in yeast. [2]

(b) Explain which product is useful in each application. [2]

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Question 14

core · 4 marks · 0610_T21_P1_20261018_Q14

Two equal masses of the same crushed fruit are treated identically except that one receives pectinase.

(a) State the expected effect of pectinase and explain why. [2]

(b) State two variables that should be kept constant. [2]

Question 15

core · 4 marks · 0610_T21_P1_20261018_Q15

A student tests a biological washing powder on identical protein stains.

(a) Describe two controls needed when comparing it with an enzyme-free powder. [2]

(b) Explain why a protease can improve removal of these stains. [2]

Question 16

supplement · 4 marks · 0610_T21_P1_20261018_Q16

Milk is treated with lactase for people who cannot digest lactose effectively.

(a) Describe the reaction catalysed by lactase. [2]

(b) Explain two advantages of using this treatment. [2]

Question 17

supplement · 4 marks · 0610_T21_P1_20261018_Q17

An aerobic fermenter produces mycoprotein using a fungus.

(a) Explain why oxygen and nutrient supply are controlled. [2]

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(b) Explain why pH and temperature are controlled. [2]

Question 18

supplement · 4 marks · 0610_T21_P1_20261018_Q18

Bacteria are modified to produce a human protein.

(a) Describe how the human gene and plasmid are prepared and joined. [2]

(b) Describe what happens after recombinant plasmids are inserted into bacteria. [2]

Question 19

supplement · 4 marks · 0610_T21_P1_20261018_Q19

A farmer grows a herbicide-resistant crop produced by genetic modification.

(a) Explain the intended benefit during weed control. [2]

(b) Describe two possible disadvantages or concerns. [2]

Question 20

supplement · 4 marks · 0610_T21_P1_20261018_Q20

A rice variety is modified to improve its nutritional content.

(a) Explain one benefit for consumers and one possible benefit for farmers. [2]

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(b) Describe two issues that should be considered before widespread cultivation. [2]

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